SOUTH AFRICAN JOURNAL OF INFORMATION MANAGEMENT, VOL 28, NO 1 (2026)

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About the Author(s)

Tshepiso P. Morema (mailto:212469408@tut4life.ac.za) (https://orcid.org/0009-0003-3707-5022)
Department of Informatics, Faculty of Information and Communication Technology, Tshwane University of Technology, Pretoria, South Africa

Tope S. Adeyelure (https://orcid.org/0000-0002-6138-4285)
Department of Informatics, Faculty of Information and Communication Technology, Tshwane University of Technology, Pretoria, South Africa

Tshinakaho R. Seaba (https://orcid.org/0000-0002-5773-887X)
Department of IT Management and Governance, Faculty of Engineering, the Built Environment and Technology (FEBT), Nelson Mandela University, Gqeberha, South Africa

Citation

Morema, T.P., Adeyelure, T.S. & Seaba, T.R., 2026, 'Cloud ERP adoption in Ekurhuleni's logistics SMEs: A conceptual model', *South African Journal of Information*

Original Research

Cloud ERP adoption in Ekurhuleni's logistics SMEs:

Tshepiso P. Morema, Tope S. Adeyelure, Tshinakaho R. Sea

Received: 01 Mar. 2026; **Accepted:** 29 Apr. 2026; **Published:**

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Abstract

Background: Logistics small and medium enterprises (SMEs) in systems. Although cloud-based enterprise resource planning (ERI

Objectives: This study aimed to develop a Cloud ERP Adoption M influencing adoption. The study was guided by the Technology–O innovation theory.

Method: A quantitative approach was used. Data were collected and logistics managers in Ekurhuleni-based SMEs. Analysis was c States, applying descriptive statistics, Pearson correlation and mu

Results: Significant relationships were observed across construc significantly influenced adoption, with 10 of 11 hypotheses suppo

Conclusion: Cloud ERP adoption among logistics SMEs is driven strategic intent rather than technical attributes.

Contribution: The study contributes by proposing an empirically digital transformation.

Keywords: cloud ERP; logistics SMEs; conceptual model; adopti

Introduction

Recent developments in information and communication technolo enterprise computing (Zamani **2022 (#CIT0045_2179)**). Traditi with evolving technologies or integrate effectively with modern di machine learning and Internet of Things capabilities to enhance e **(#CIT0012_2179)**). These developments are especially relevant stimulating economic activity, enabling market access and suppor face rising operational demands driven by e-commerce, increased **(#CIT0040_2179)** ; Yan, Wang & Chen **2023 (#CIT0044_2179)** fragmented systems, resulting in delays, data inaccuracies, highe). Modern cloud ERP systems offer a pathway to overcome these decision-making (Nagahawatta, Singh & Kumara **2024 (#CIT002** paid limited attention to the specific conditions and challenges ex literature focuses on large organisations or general SME contexts technological factors that influence adoption within logistics-inter Monticelli **2024 (#CIT0006_2179)**). Small and medium enterpri

Management 28(1), a2179.
https://doi.org/10.4102/sajim.v28i1.2179
(https://doi.org/10.4102/sajim.v28i1.2179)

technology (IT) expertise, financial pressures, organisational resi
(#CIT0036_2179)). These factors shape their readiness for clou
circumstances. Understanding these dynamics is essential, as clo
on local infrastructure and strengthen long-term sustainability (A
(#CIT0013_2179)). Despite these recognised opportunities and
contexts or specific national settings, with limited empirical resea
influence cloud ERP adoption within logistics-focused SMEs in Eku
logistics sector, highlighting a contextual and empirical gap that r
(#CIT0018_2179) ; Nagahawatta et al. **2024 (#CIT0026_2179)**

Research problem

The logistics sector in Ekurhuleni acts as a substantial contributor
SMEs continue to face a range of operational and strategic challe
implementation costs and persistent concerns over data security.
and competitiveness, SMEs in Ekurhuleni remain hesitant becaus

Existing research on cloud ERP adoption has largely focused on la
dynamics that characterise logistics SMEs in emerging economies
(#CIT0002_2179)) examine adoption in larger corporate setting
and infrastructural challenges experienced by smaller logistics fir
SMEs without clear guidance on how to navigate adoption barrier
to transform logistics SMEs by integrating core business functions
coordination into a single, efficient platform. Additionally, cloud E
reducing the financial and technical burden typically associated w

Scholars, including Manigandan and Raghuram (**2024 (#CIT0021
(#CIT0006_2179)**) and Shahadat et al. (**2023 (#CIT0038_2179**
cloud ERP adoption among SMEs, especially within emerging econ
the adoption of cloud ERP in logistics SMEs within Ekurhuleni, offe
implementation. Consequently, the study's research objectives ar
assess the challenges faced by logistics SMEs and evaluate the p
ensures the development of actionable insights that can guide bo
objectives were structured to provide a clear pathway for develop
Ekurhuleni's logistics SMEs.

Research aim and objectives

The study aimed to develop a model for the adoption of cloud ER
objectives:

- *To assess existing resource-planning challenges faced in the absen*
- *To identify factors influencing the adoption of cloud ERP*.
- *To determine the influence of identified factors on adopting cloud*
- *To design a model for adopting cloud ERP in Ekurhuleni logistics S*

Having outlined the aim and objectives, the study required a stro
relationships among variables and support the eventual design of
technology-adoption frameworks that offer insight into how organ

Related theoretical frameworks

Related literature

Recent studies highlight growing interest in cloud ERP adoption a
(**2022 (#CIT0011_2179)**) emphasise the need for more context
reflect sector-specific demands. Tongsuksai, Mathrani and Weeras
Zealand SMEs, yet their findings do not address the realities of de
in South African SOEs (Nakeng, Mokwena & Moeti **2021 (#CIT00**
organisational readiness and leadership support, but these studie
(**2023 (#CIT0023_2179)**) and Christiansen et al. (**2022 (#CIT0**
models (CEAMs), noting that existing studies often reflect limited
study responds by proposing a model tailored to the operational,

Related frameworks informing the conceptual model

Building on prior work by Razzaq et al. (**2021 (#CIT0035_2179)**
the Iacovou model to conceptualise cloud ERP adoption in Malays
Ekurhuleni. Diffusion of innovation contributes constructs such as
technological perceptions influencing adoption decisions. Technol
readiness, financial capability, perceived benefits and environmer
emphasises the importance of leadership support, effective comm
the Iacovou model reinforces SME-specific determinants, includin

(#CIT0039_2179)). Collectively, these theoretical lenses inform view of cloud ERP adoption, demonstrating why a comprehensive

Tongsuksai et al. (**2023 (#CIT0041_2179)**) merge TOE and Unif and user acceptance. Technology Enterprise User Trust (TEUT) er influence and facilitating conditions while highlighting the critical This integration ensures that the proposed model accounts for bo **(#CIT0023_2179)**) Cloud ERP Implementation and Technology I Model (ISSM) success model, emphasising infrastructure, compat layered understanding of cloud ERP adoption in developing-count and organisational readiness in logistics contexts. Together, these theory contributed the technological perception constructs – syst contributed organisational constructs – top management support edge and vendor support. Technology Enterprise User Trust contr readiness. Cloud ERP Implementation and Technology Integratior developing-country SME contexts. Together, these frameworks su contributes a different but complementary view: technology read and user acceptance. When combined, they provide a more comp decisions are influenced by limited resources, operational deman reflects the practical realities faced by logistics SMEs while remai

Quantitative content analysis

Following Parry's (**2020 (#CIT0032_2179)**) approach, a quantita Systematic Reviews and Meta-Analyses (PRISMA)-style bibliomet databases using predefined keywords related to cloud ERP adopti peer-reviewed empirical studies were retained. Frequency analys adoption factors, which informed the development of the concept (95), security (90) and complexity (86) were the most dominant and system usability. Organisational factors such as top manager leadership commitment and resource readiness as critical enable (50) and organisational structure (60), pointing to the significanc

TABLE 1: (https://sajim.co.za/index.php/sajim/article/viewFil

To validate and extend the literature-derived frequency counts, a corresponding survey construct measured in the primary study. F = 123) using IBM SPSS Version 31.0.0.0: (1) the composite mea agreed or strongly agreed (≥ 4 on the five-point Likert scale). Th **(#SAJIM-28-2179-E2)** and **Equation 3 (#SAJIM-28-2179-E3)** :

Equation 1 (#SAJIM-28-2179-E1) : Composite mean score (M)

$$M_i = (1 / (n \times k)) \times \sum_{j=1}^{k} \sum_{j=1}^{n} x_{ijx} \quad \text{[Eqn 1]}$$

where M_j = composite mean for construct i, n = number of respo respondent j on item k of construct i (scale: 1–5).

Equation 2 (#SAJIM-28-2179-E2) : Standard deviation (SD)

$$SD_i = \sqrt{[(1 / (N-1)) \times \sum_{j=1}^{R} (x_{ij} - M_i)^2]} \quad \text{[Eqn 2]}$$

where SD_j = standard deviation for construct i, N = total number individual item response r for construct i and M_j = composite mea

Equation 3 (#SAJIM-28-2179-E3) : Percentage agreement (% a

$$\%Agree_i = (f_i (\geq 4) / N_i) \times 100 \quad \text{[Eqn 3]}$$

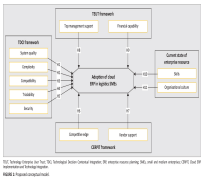
where $\%Agree_j$ = percentage of item responses rated ≥ 4 for cor for construct i, N_j = total number of individual item responses for response, giving N_j = 368). To illustrate **Equation (3) (#SAJIM-** Therefore, %Agree (Security) = (300 / 369) × 100 = 81.3%, rou respondents whose average across the three security items was as it better reflects the respondent's overall position on a constru resulting descriptive statistics alongside the literature frequency scholarly literature emphasises and how logistics SME practitione convergence between the bibliometric signal and the primary dat academic emphasis.

TABLE 2: (https://sajim.co.za/index.php/sajim/article/viewFil
(*N* = 123).

Table 2 (#T0002_2179) reveals a high degree of convergence b factors recorded composite mean scores above 4.0 on the five-p four or higher (computed using **Equation (3) (#SAJIM-28-2179** convergence provides dual validation for the factor selection und adoption literature but also are practically perceived as significan second-highest literature frequency count (90) and the second-h **28-2179-E1)** and **Equation 2 (#SAJIM-28-2179-E2)**), with 88.5 adoption enabler and a potential barrier. System quality, despite mean score (*M* = 4.242, SD = 0.894) and the highest agreement practical importance to system performance than the broader aca logistics SMEs on real-time data accuracy and system reliability. mean score (*M* = 4.097, SD = 0.957), consistent with the patter decisive adoption determinant in cloud ERP contexts relative to o in the construct selection and reinforce the theoretical coherence conceptual model development.

Conceptual model

The proposed framework includes technological, organisational a Ekurhuleni in adopting cloud ERP solutions. To achieve this, the n conceptual insights from existing models such as TDCI, TEUT and efficiency. Specifically, TDCI emphasises technological attributes compatibility, trialability and security, with a specific emphasis on emphasises the organisational aspects influencing technology ado management support and organisational readiness as focal const environmental factors, particularly vendor support and competitiv gain an edge. The resulting CEAM integrates organisational, envi logistics SMEs. The constructs encompass system quality, comple management support and organisational culture, with cloud ERP model for cloud ERP adoption'.



(https://sajim.co.za/index.php/sajim/article/viewFile/2179/3

Proposed model and hypotheses formulation

H1: System quality will affect cloud ERP adoption in Ekurhuleni lo **(#CIT0005_2179)**).

H2: Complexity will affect cloud ERP adoption in Ekurhuleni logist **(#CIT0022_2179)**).

H3: Compatibility will affect cloud ERP adoption in Ekurhuleni log **(#CIT0010_2179)**).

H4: Trialability will affect cloud ERP adoption in Ekurhuleni logisti

H5: Security will affect cloud ERP adoption in Ekurhuleni logistics

H6: Competitive edge will affect cloud ERP adoption in Ekurhulen

H7: Vendor support will affect cloud ERP adoption in Ekurhuleni l **(#CIT0011_2179)**).

H8: Top management support will affect cloud ERP adoption in Ek

H9: Financial capability will affect cloud ERP adoption in Ekurhule

H10: Skills will affect cloud ERP adoption in Ekurhuleni logistics S

H11: Organisational culture will affect cloud ERP adoption in Ekur

Overall, the theoretical and conceptual models discussed provide
SMEs. They guide the development of the proposed model and in
methodology used to investigate these factors.

Research methods and design

This study adopted a positivist, quantitative research design, und
against empirical data (Saunders, Lewis & Thornhill **2023 (#CIT(**
established practice in information systems adoption research, as
time (Vomberg & Klarmann **2021 (#CIT0043_2179)**). A mono-m
analytical rigour. The use of a quantitative approach is appropriat
defined adoption constructs rather than explore subjective meani

Data collection

Data were collected through a structured, closed-ended question
email and WhatsApp to middle and senior managers, including IT
electronically through Google Forms, ensuring efficient data colle

Questionnaire structure

The questionnaire included three main sections:

- Demographic information capturing organisational size, participant
- Knowledge screening assessing familiarity with ERP and cloud com
- Measures of study constructs including system quality, complexity
  support, skills, organisational culture, competitive edge and cloud

All items were measured on a five-point Likert scale (1 = strongly
technology-adoption research, ensuring content validity. Data we

Sampling techniques

The process of choosing a subset of a specified population to part
sampling (Nanjundeswaraswamy & Divakar **2021 (#CIT0028_21**
operations managers, IT managers and IT personnel. This compo
decision-making and system usage. The study population compri
focused SMEs operating in Ekurhuleni Municipality, Gauteng. Part
industry specialists within logistics SMEs, to be directly involved i
regarding technology adoption. Respondents were identified thro
Business Chamber, logistics industry directories and professional
Airport. Snowball referrals within participating organisations supp
sizes. To calculate the required sample size, this study used a sim
population an identical possibility of selection, which helps reduce

According to Hossan and Alhasnawi (**2023 (#CIT0017_2179)**), t
about a population. In contrast, the population size denotes the e
(#CIT0008_2179)). The margin of error indicates the extent to
Dessel **2013 (#CIT0042_2179)**).

Based on a preliminary investigation conducted within the study's
180 individuals within the logistics SMEs in the Ekurhuleni munici
established, based on a 5% margin of error and a 95% confidenc

Data analysis

Data analysis was conducted using IBM SPSS Version 31.0.0.0. A
values, response completeness and univariate outliers to ensure
assessed for each construct using Cronbach's alpha, with a minim
Thirdly, construct validity was evaluated through exploratory fact
variance extracted (AVE ≥ 0.50) for convergent validity and the F
(#CIT0016_2179)). Fourthly, Pearson's bivariate correlation ana
11 independent constructs and the dependent variable (cloud ER
study hypotheses by estimating the standardised regression coef
each construct's unique contribution to explaining variance in adc

Ethical considerations

Ethical clearance was obtained from Tshwane University of Techn
Participation in the study was entirely voluntary, and informed co
confidentiality were strictly maintained throughout the study. No
from the study at any time without penalty. All data were securel

Results

This section presents the results of the survey conducted among participants, reliability and validity of the measurement instrume hypotheses.

Table 3 (#T0003_2179) summarises the demographic profile of

TABLE 3: (https://sajim.co.za/index.php/sajim/article/viewFil

Overall, the demographic profile demonstrates a diverse distribut base for analysing cloud ERP adoption within logistics SMEs.

Reliability and validity of the study

Reliability analysis is essential for determining the internal consis **(#T0004_2179)**) was used to assess construct reliability, in line Moumane, Benabdellah and Rafik (**2023 (#CIT0024_2179)**), all thresholds (α ≥ 0.70) recommended by Nanjundeswaraswamy ar Bernstein (**1994 (#CIT0030_2179)**) support these benchmarks.

TABLE 4: (https://sajim.co.za/index.php/sajim/article/viewFil

All constructs exceeded the acceptable threshold of 0.70, indicati 0.903) demonstrated the highest reliability, while system quality differences between raw and standardised Cronbach's alpha valu internal coherence of the measurement tool and validates its suit and increases confidence that observed relationships, such as the rather than instrument inconsistencies (Kennedy **2022 (#CIT001**

Validity of the constructs

Construct validity was assessed through EFA and confirmatory fa (≥ 0.50) on their intended constructs, confirming item–construct assessed through AVE, with all constructs achieving AVE ≥ 0.50, (Cheung et al. **2023a (#CIT0009_2179)**). Discriminant validity construct's AVE exceeded its correlations with other constructs, c

Pearson's correlation of the constructs

Pearson's correlation analysis was conducted to assess the direct **(#T0005_2179)** , all correlations were positive and statistically si organisational and human factors. Strong correlations were obse skills and top management support (r = 0.841, p < 0.01), sugge readiness. Moderate correlations were also noted, such as trialab 0.547, p < 0.01), highlighting the interconnected nature of techn

TABLE 5: (https://sajim.co.za/index.php/sajim/article/viewFil

Regression analysis

Multiple regression analysis was conducted to assess the influenc found to be statistically significant. Several predictors showed str = 0.466, p < 0.001), underscoring the critical role of leadership organisational culture (β = 0.235, p = 0.021), edge (β = 0.213, existing literature that highlights the importance of managerial, c

In contrast, several constructs did not show statistically significar compatibility, system quality and vendor support. This trend is co and strategic considerations often outweigh purely technical char relevant, they are less influential than internal organisational dyn

Hypotheses testing

Twelve hypotheses (H1–H12) were formulated to assess the influ logistics SMEs. Based on the regression outputs presented in **Tab** (H8) was not supported, and H12 represents the dependent varia

TABLE 6: (https://sajim.co.za/index.php/sajim/article/viewFil

TABLE 7: (https://sajim.co.za/index.php/sajim/article/viewFil

The supported hypotheses indicate that several technological, org
adoption. These include system quality (H1), complexity (H2), co
capability (H7), organisational culture (H9), competitive edge (H1
these constructs contribute meaningfully to the variance explaine
construct did not have a statistically significant influence on the r
predictor but served as the outcome variable for all hypotheses. (
technological and strategic factors contribute to cloud ERP adopti
regression analysis.

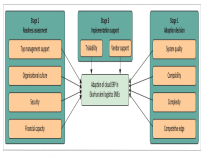
The findings presented in **Table 7 (#T0007_2179)** provide a bas
SMEs. The next section discusses these results in relation to exist

Discussion

This study set out to develop an empirically grounded CEAM for l
specific adoption frameworks in developing-economy settings. Th
by organisational and strategic conditions rather than by technolo
centric adoption narratives and repositions cloud ERP adoption as
provides nuanced and differentiated insights into adoption detern
predictors. System quality (H1) showed a modest positive influer
they are insufficient without managerial alignment and strategic
continuing role as a major inhibitor in SMEs with limited change
suggesting that logistics SMEs prioritise strategic alignment and
particularly distinctive and noteworthy finding is the negative rela
this result suggests that poorly managed or inadequately structu
rather than encourage adoption. This finding extends existing ad
environments and reinforces the need for governance even durin
of adoption. Top management support (H6) was the strongest pr
and sustaining implementation momentum. Security (H5) also de
protection, regulatory compliance and system trust in logistics en
moderate but meaningful influence, confirming SME cost sensitiv
(H8) were not supported, suggesting a structural shift in adoptio
reduced the importance of in-house technical expertise. Environn
(H9) and competitive edge (H10) both positively influenced adop
digital transformation. Although vendor support (H11) was statis
compensate for weak internal leadership or governance structure
adoption) and affirm the relevance of the TOE framework within t
organisational readiness and leadership commitment. The implica
alongside the value and contribution of the CEAM.

Value and contribution of the cloud enterprise resource planning

The proposed CEAM (**Figure 2 (#F0002_2179)**) makes a substa
transformation practice. Four dimensions of its uniqueness merit



(https://sajim.co.za/index.php/sajim/article/viewFile/2179/3

Firstly, the CEAM is the only empirically validated, sector-specific
municipal context. Existing adoption models such as TDCI (Razza
and TEUT (Bayag & Madimabe **2024 (#CIT0007_2179)**) were de
and do not account for the operational realities – load shedding v
limited ICT capital – that characterise logistics SMEs in Ekurhuler
adoption frameworks that reflect sector and geography (Christiar

Secondly, the CEAM repositions cloud ERP adoption as a leadersh
decision. The finding that top management support (β = 0.466) e
technology-centric framing that dominates the cloud ERP adoptio
implementation conversations from feature demonstrations to lea
transformation incentive programmes that address executive awa
non-significance of internal IT skills, the CEAM empirically confirn
Software as a Service (SaaS) delivery has materially lowered the
logistics SMEs in developing economies where ICT skills shortage
sufficient justification for non-adoption. Fourthly, the CEAM provi

managers, it prescribes a leadership-first adoption readiness seq
culture before committing capital to software implementation. Fo
strategies should prioritise security assurance, competitive differe
For policymakers and development agencies, including the Ekurh
it highlights the importance of executive digital literacy programr
preconditions for sustainable cloud ERP uptake in the logistics se

Recommendations for future research

Future studies should expand the geographical scope to include a
Community (SADC) countries to enable comparative analysis and
states dedicated to socioeconomic cooperation, political integratic
mixed-method approaches, incorporating interviews or case stud
Further research may validate and refine the CEAM across other s
frameworks for SMMEs based on organisational size and digital m
evolves over time and to assess the sustained impact of cloud ER
the determinants of cloud ERP adoption among logistics SMEs in
validated, context-specific adoption framework. Using a quantitat
practitioners through Pearson's correlation and multiple regressio
determinants and cloud ERP adoption intention.

The findings establish that cloud ERP adoption in this context is d
support, organisational culture, security perceptions, competitive
significance of skills availability signals a structural shift in the ad
architectures have materially reduced the in-house technical expe
resource-limited SMEs.

From a theoretical standpoint, the study reinforces and contextua
specific setting. Notably, the negative trialability finding challenge
literature: in SME environments with limited governance capacity
finding merits further investigation in future studies. Practically, t
adoption in the logistics sector. Small and medium enterprise ma
preconditions. Vendors should recalibrate implementation strateg
the Ekurhuleni Metropolitan Municipality, should invest in executi
cloud adoption. Future studies should test the CEAM across other
manufacturing and retail and should employ longitudinal designs
effects.

Conclusion

This study analysed cloud ERP adoption among logistics SMEs in
highlighted that SMEs without cloud ERP face significant operatio
benefit from improved integration and decision-making. The stud
the 11 hypothesised factors significantly influenced adoption. Top
significant factor. These findings informed the development of the
ERP adoption.

Acknowledgements

This article is based on research originally conducted as part of T
SMEs: A Conceptual Model', submitted to the Department of Infor
Technology, in 2026. The thesis is currently unpublished and not
thesis. The study was reworked, revised and adapted into a journ
published or disseminated and complies with ethical standards fo

Competing interests

The authors declare that they have no financial or personal relati

CRediT authorship contribution

Tshepiso P. Morema: Conceptualisation, Data curation, Formal an
draft. Tope S. Adeyelure: Supervision, Writing – review & editing.
article, contributed to the discussion of results, approved the fina
findings.

Funding information

This research received no specific grant from any funding agency

Data availability

The data that support the findings of this study are available fron

Disclaimer

The views and opinions expressed in this article are those of the
official policy or position of any affiliated institution, funder, agen
and content.

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